



Makerspace and Prototyping Safety Guideline

Health and Safety Considerations for the Management of Makerspaces, including Prototyping and Product-Development Spaces

Makerspaces, including any prototyping and product-development spaces, are facilities that encourage collaboration, hands-on learning, experimentation, and creation, including do-it-yourself projects using hand and power tools. The types of activities and equipment may overlap with other facilities, such as machine shops where activities are carried out by professional or technical staff; however, unlike those settings, the activities in makerspaces tend to be self-guided by the user.

The types of activities, equipment, machinery, and tools at each makerspace facility will vary depending on the objectives of the space. The purpose of this document is to articulate some high-level considerations for departments that are planning for, or are currently operating, such a space and should not be interpreted as an exhaustive or comprehensive list. Maker and prototyping spaces need to prioritize safety for all users, both internal to the university (employees, students) and external (visitors, community members, collaborators) as applicable.

While fabrication device use, including 3D printers, is common for research and academic purposes, it is important for departments to understand that there are certain activities – such as replicating patented objects, mass-producing items for commercial sale, or printing weapons or medical devices – that can give rise to significant legal and safety issues.

It is expected that departments will review this document and carefully consider the manner in which their makerspaces are being used by the University community. Departments should contact EHS (ehs.office@utoronto.ca) if they have any concerns or require assistance.

For all equipment that is built or significantly repurposed, modified, or altered for use in a research *lab* where equipment design or modifications have been directed by U of T staff or students, whether or not the equipment is built at U of T, please refer to the [Guidelines on University Designed/Built Equipment Safety](#).

For any questions regarding core facilities, defined as any research facilities across the institution's three campuses offering specialized research services or equipment to the U of T community, please visit the [VPRI Institutional Core Facilities webpage](#).

Design Considerations

Layout and Space Usage

- Adequate space for movement, machinery, and materials-handling (ergonomics).
- Storage areas are designed to provide adequate space for personal protective equipment (PPE), tools, equipment, and materials (including hazardous chemicals, storage of compatible chemicals).
- Waste management should include proper sorting, storing and organizing including off-cuts and shavings. When applicable, a designated area for hazardous, flammable, explosive and toxic waste should be allocated.
- Workstations for prolonged use are located away from printers, including 3D printers, and other emissions.
- Signage and labels are posted with thorough and applicable safety considerations. Any integration of any new equipment into the building envelope requires consultation with the property manager of the building (e.g. anchoring storage shelves to the wall, installing a window AC unit, connecting power tools to power, etc.).

Engineering Controls

- Adequate room ventilation based on activities, exhaust (including local exhaust ventilation), and filtration when applicable (e.g., chemicals, welding fumes, wood dust, heat generated from equipment, etc.).
- Include fume hoods or other local ventilation in the design of the makerspace when dealing with volatile or hazardous substances to reduce exposure to chemicals.
- Adequate lighting, including task lighting over work areas and benches for small parts assembly.
- Electrical needs shall be adequate for the Electrical Safety Association (ESA) certified equipment and follow the EHS [Electrical Equipment Guidance](#).
- Incorporate noise controls such as enclosures, baffles, purchasing quieter equipment, etc. to reduce noise from equipment, activities, space infrastructure, etc.
- Include dust controls when items are drilled, cut, sanded or polished and fume controls for soldering, welding, etc.
- Select furniture with ergonomic features such as adjustability (worksurface), lumbar support, armrests, footrests, and tilt angles (chairs). Design for flexibility, using furniture that can be easily moved and reconfigured to suit changing makerspace needs. Look for furniture that can be easily adjusted to fit the height, weight, and posture of different users. Use finishes that can withstand spills and repeated cleaning, e.g., avoid absorbent or flammable materials.

Emergency Preparedness and Security

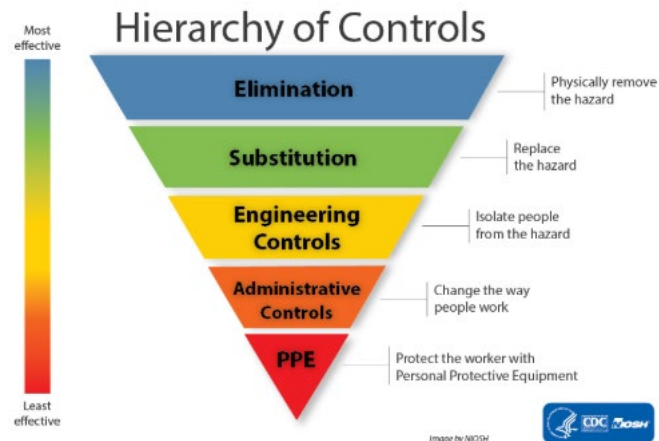
- Incorporate fire protection measures and equipment (at a minimum, smoke and fire detection and alarm systems), including extinguishers appropriate for the hazards
- Provide clearly defined emergency egress routes. Floor markings are used, where applicable, to demarcate between hazard or work zones and walkways.

- Provide eyewash and emergency deluge showers, hand washing sink(s) based on the required activities.
- Ensure controls are in place to prevent unauthorized access.

Health & Safety Plans

Departments are responsible for supervising their makerspaces, and for developing and implementing health and safety measures that are applicable to the particular facility.

1. Develop, review, and implement facility-specific health and safety plans, training, procedures (*see guidance below)
2. Conduct a hazard assessment and develop standard operating procedures (SOPs) for equipment, activities and substances. Refer to manufacturer's instructions, safety data sheets (SDS), and industry best practices.



Source: U.S. National Institute for Occupational Health and Safety (NIOSH)

3. Develop mandatory training for staff and users (e.g., faculty, staff, students, visitors, external users etc.), which includes facility-specific training (*i.e.*, the exact types of equipment, protective measures, emergency procedures, etc. in the facility). Maintain records of training/orientation (e.g., dates, attendees, topics covered, etc.) and provide refresher training on a regular basis (e.g., annual). In general, facilities users will be expected to use equipment/tools provided/maintained by the makerspace and not to store items at the makerspace. This is to reduce the risk of injury as outside tools/equipment may not be built to Canadian standards or safety requirements, be maintained less than recommended, be modified outside the intended use and may have potential hazards from storage (e.g. in the event of fire, flood, spills, incompatible chemicals or other emergency). In rare circumstances where outside items are permitted for specialized activities, additional safety measures will be needed. Please contact EHS for additional recommendations and assistance.
4. In most cases, it is expected that these spaces operate during normal business hours of the building. This is to ensure staff is available to monitor activities and to provide assistance, especially in case of an emergencies. In rare situations where spaces are open during non-business, additional procedures are required (e.g. rotating on-call schedule for staff, limiting activities to low risk ones, arranging for Campus Safety or patrols to check on the space hourly, check-in and check-out procedures, etc.). Please contact EHS for assistance.
5. Develop rules and a process for reviewing and approving projects to avoid inappropriate objects from being made (e.g., weapons, weapon parts, medical devices, items that can cause physical or emotional harm, etc.).

6. Develop a user agreement that requires users to agree to the rules, guidelines and procedures and training for the facility. The user agreement should specify that the facility has the right to reject a project and the user must not proceed with that project. Please refer to Appendix A for a sample template. If a department is interested in developing a waiver, , contact Risk Management and EHS to discuss the use of waivers.
7. Identify a shop safety coordinator, or responsible person to provide supervision, advice, training, and enforce safety rules for all users. Ensure workplace supervisors are knowledgeable about their roles and responsibilities (Resource - "[I am a Supervisor](#)").
8. Control access so that only trained, authorized persons can use the space/equipment.
9. Conduct regular inspections, including checking first aid kits and fire extinguishers. (Resource: [JHSC Workplace Inspection Checklist](#)." Departments are responsible for incorporating facility/equipment-specific considerations into their checklist.)
10. Schedule and conduct routine maintenance of machinery, equipment, and tools.
11. Contact EHS for context specific review risk assessment/information (e.g. personal projects vs. curriculum-based activities, types of equipment, etc).
12. Review and update the above on a regular basis (e.g. annually) and communicate changes/updates. Keep copies of all relevant documentation and develop a document retention schedule.
13. Advise EHS (ehs.office@utoronto.ca) of any makerspace on-campus.

*Examples of topics to include in a health and safety plan:

- Any required training. For information on EHS courses, please refer to EHS Training matrices for [Facilities & Admin](#) staff and [Lab Personnel](#).
- Checks/inspection/calibration of equipment before use. Procedures on how to report any observed damages, deficiencies or concerns.
- Frequency of housekeeping measures (e.g. declutter, clear egress routes, clean up dust and debris, etc.).
- Emergency procedures, including location/access to emergency equipment (e.g. first aid kit, emergency deluge shower, eyewash, fire extinguisher, spill kit, etc. when applicable), spill procedures and contact information based on materials used or allowed in the facility (e.g. Makerspace staff, Campus Safety, Environmental Protection Services).
- Location of important health and safety information and equipment (e.g. manufacturer manuals, Safety Data Sheets (SDSs), SOPs, personal protective equipment (PPE), lock-out tag-out equipment, etc.).
- Appropriate waste disposal procedures for both hazardous and non-hazardous waste. (Resources: EPS [Bucket List](#) poster and [EPS](#) webpage.).
- Report all injuries and near-misses to the supervisor/Makerspace U of T contact who will then submit an [Incident/Accident eForm](#).

Makerspace staff members and facility users

- Participate in applicable training/orientation.
- Inspect machinery, equipment and tools before each use. Do NOT use any damaged machinery, equipment and tools and immediately report the hazard to the facility contact or their workplace supervisor.
- Operate all machinery, tools and equipment safely and in compliance with training, applicable SOPs and manufacturer information/manuals.
- Use PPE and other safety equipment as required.

- Practice good housekeeping and hygiene.
- Report health and safety hazards, concerns and injuries to the facility contact or their workplace supervisor.

Machinery, Equipment, and Tools

- Ensure equipment safety through safeguarding methods, air filters or exhaust ventilation, [lockout/tagout procedures](#), proper securing of non-portable equipment/machinery, regular calibration and/or preventative maintenance and inspection of equipment/machinery. Maintain a log of maintenance or inspection performed.
- Develop a process for reporting damaged machinery, equipment and tools and taking them out of service until repairs are completed/replacement is available.
- In addition to general ventilation, where applicable, provide local ventilation and/or equipment with dust collection has been considered based on the activity (e.g. wood dust, welding).
- Consult with [Fire Prevention](#) with regards to hot work such as welding, cutting with a torch, using equipment to melt materials, etc.
- Where applicable, ensure machine guarding and emergency stop buttons are in place to protect users. Power tools and milling machines controlled by a computer program and robotic equipment should be equipped with the manufacturer's certified guarding. Where applicable, consult with the manufacturer prior to making any changes to the guarding or to the equipment/machinery/tools.
- All electrical equipment must meet Canadian national safety standards and provincial codes and be certified by an accredited certification body such the Canadian Standards Association (CSA), the Underwriters Laboratories of Canada (ULC), etc. (Resource: [Electrical Equipment Guidance](#))
- Resources:
 - [Machine Safety Guidelines](#)
 - [Soldering Guidelines](#)
 - [Welding Guidelines](#)

3D Printers

- Only approved or comparable materials (manufacturer recommended) should be used in the 3D printer.
- Ensure appropriate ventilation applicable to the emissions produced (e.g. ultrafine particles).
- Consider using feedstock that produces less toxic emissions (e.g. ABS, when compared to PLA, requires higher temperatures and results in more toxic byproducts such as hydrogen cyanide).
- Consider the power of moving heads and the requirements for safeguarding (e.g. some 3D printers with a powerful moving head will push objects (hand) out of the way).
- Consider how scaffolding will be removed and waste disposal considerations.
- Develop procedures for cleaning 3D printers, for removing items safely from the printers and procedures to follow in case of failed print (e.g. burns from hot nozzle and extrusion heads, cuts from tools, shards or sharp edges, tie bag hair or baggy clothing when exposed

to moving parts, do not attempt to manually remove malfunctioning parts during a failed print and instead, use the control panel or cancel the print job).

- Some 3D printers use UV lights. If exposure to UV light may occur, consult with EHS regarding eye protection.
- Resource: <https://www.canada.ca/en/employment-social-development/services/health-safety/prevention/3d-printers.html> The first section of this webpage provides helpful guidance regarding 3D printers.

Chemical Safety

- If work involves Controlled Products under WHMIS, take WHMIS training. For information on EHS courses, please refer to EHS Training matrices for [Facilities & Admin](#) staff and [Lab Personnel](#).
- Ensure appropriate spill kits are available, and staff are trained in their use. For large spills, contact Environmental Protection Services (416-978-7000).
- Develop procedures for the storage of chemicals (e.g. approved flammable liquid cabinet, appropriate separation of incompatible materials).
- Ensure chemical Safety Data Sheets (SDSs) are in [HECHMET](#). (Resources: [Chemical Storage Flowchart](#) and [Chemical Storage Table](#).)
- Where available, eyewash is flushed weekly (logged) by the makerspace and emergency showers are tested annually by Facilities & Services. If a chemical fume hood is used, an annual certification by EHS is required (refer to the sticker on the sash or contact ehs.office@utoronto.ca).

Prohibited Items

At minimum, the following items are prohibited:

- Weapons and weapon parts
- Guns and gun parts

It is illegal to manufacture or possess any firearm — including a 3D-printed firearm or certain components of a firearm — without the appropriate license issued by a Chief Firearms Officer. For more information, Please refer to: <https://www.canada.ca/en/public-safety-canada/news/2024/08/new-firearms-licensing-requirements-to-increase-public-safety-coming-into-force.html>.

Medical Devices

There are regulations regarding licensing of medical devices produced by 3D printing. It is the responsibility of all facilities users to thoroughly review all applicable regulations, legislation, guidance or directive and comply with them accordingly. Some resources have been included in the below. Please also discuss your project with the maker space directly.

- Legislation and Guidelines – Medical devices: <https://www.canada.ca/en/health-canada/services/drugs-health-products/medical-devices/legislation-guidelines.html>
- Safe medical devices in Canada: <https://www.canada.ca/en/health-canada/services/drugs-health-products/medical-devices/activities/fact-sheets/safe-medical-devices-fact-sheet.html>.

Please contact EHS for any questions.

Additional Resources

- Please refer to the main page ([EHS policies and procedures](#)) for key policies and procedures, including but not limited to the following:
 - [Chemical Storage Table \(PDF\)](#)
 - [Eyewash and Shower Needs Assessment Flowchart \(PDF\)](#)
 - [Laboratory PPE Assessment Tool \(PDF\)](#)
 - [Laser Safety](#)
 - [Respiratory Protection Program \(PDF\)](#)
 - [Working Alone Guidelines \(PDF\)](#)
 - Working alone may require additional measures such as limiting the hours where working alone is permitted, prohibiting higher risk activities/locking up higher risk machinery during working alone periods, developing a sign-in procedure and/or arranging for regular patrols of the area. Contact EHS for further information.
- Fire Extinguisher Training on Campus: [Fire extinguisher training - Facilities & Services - University of Toronto](#)
- First Aid Training: [First Aid Training - Environmental Health & Safety](#)
- For staff who develop SOPs, EHS303 online Job Safety Analysis training (EHS303) is recommended: <https://ehs.utoronto.ca/training/my-ehs-training/>. This course teaches participants on how to use the [Job Safety Analysis](#) tool to break down tasks and identify hazards and controls to reduce/eliminate the hazard. Examples of templates can be found from the University of Nevada [Basic Makerspace safety](#) and Harvard University [Makespace Safety](#) webpages. Each department must develop their own procedures applicable to their facilities.
- [University of Toronto Preparedness](#)

Appendix A – User Agreement Template Language

Instructions: The language below contains basic expectations that the Facility's user is required to follow including facilities dedicated to academic course work/projects. Departments may add additional instructions, expectations, etc. that are relevant to their facilities. Departments should also implement a process for re-signing the user agreement at an interval that is applicable to their operations but at minimum, at least annually (e.g. every September) to refresh Facility Users of their obligations.

Ongoing compliance with these guidelines is mandatory for continued access to the facility. The Facility is a shared space and users must conduct themselves in a manner that does not negatively impact the health and safety of all individuals who use or work in the space. Additional guidelines may be communicated to the facility user at any time as issues arise and when management of the Facility deems necessary to ensure safety and that all activities are lawful and adhere to appropriate intellectual property laws.

The following guidelines must be adhered to when accessing and performing any activities at the Facility [insert Name of Facility]:

1. All activities by the Facility user must adhere to all applicable laws, including but not limited to intellectual property laws.
2. Use of the Facility to create, produce or alter any items that are illegal (e.g. weapons or weapon parts, drug paraphernalia), unsafe, harmful, or obscene, as determined by Facility management in its sole discretion, is strictly prohibited.
3. Use of the Facility for commercial purposes may be declined. [Where applicable, Facility should include other prohibitions, e.g. The Facility can only be used for course projects that have been approved and not for personal projects, etc.]
4. The Facility user will be responsible for any damage to equipment and facilities as a result of negligence and/or non-compliance with the facility's policies and procedures.
5. The Facility user will respect the reservation time and perform activities within the scheduled timeframe.
6. The Facility user will report any health and safety issues, incidents and injuries to the designated Facility contact immediately. [insert Name of Designated contact and their contact information]
7. The Facility user will complete all health and safety training required by the Facility prior to commencing activities.
8. The Facility user will comply with all applicable regulatory requirements and University of Toronto policies and procedures, including without limitation all EHS [guidelines](#) and Facility-specific health and safety requirements and procedures at all times: [Insert links to facility-specific guidelines/procedures. Departments may also insert some specific points that they wish to emphasize into the agreement]

[If User Agreement is greater than 1 page, insert an initial box for the Facilities User to initial on the bottom of each page].

Initials	
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